

We have a luxury buyer **with** an **unlimited** budget

Willing **to** have:

**Large or** locally dominant house

Big **space** (tennis, golf)

Historic

**No** waterfront

# Retrieving data from the database through psycopg2

Left join on house ids

And saved the data as .csv file

# Data cleaning

No strange characters (e.g. “\_ ! ,” or space) in column names

No capital letters

Some extra 0s in the end of some values in yr\_renovated: removed by dividing those vals by 10

No duplicated columns

Dtype of “view”: bool

Dtype of “date”: Timestamp

Some float (like 0.5 or 1.5) in floor/num\_bathrooms/num\_bedrooms; so we left the dtype of those untouched (float)

Missing values in sqft\_basement (2.09%), waterfront (11.07%), and yr\_renovated (17.82%)

Added a column to the data: house age = 2026-yr\_built

# Fill nan s in waterfront

## Approach

1. We used rows where waterfront is known (0 or 1) as training data
2. And (lat, long) as features
3. Then we train a KNeighborsClassifier to predict missing (NaN) values in waterfront
4. We chose a reasonable number of neighbors (10) to train the data
5. Also, we used weights='distance' so closer neighbors matter more

**Accuracy = 0.993**

- We *could* use zipcode to impute waterfront, and it would be actually a very sensible baseline. But zipcode is a **coarser signal** than (lat, long)

# Fil NaNs in yr\_renovated

We first need to check if there is a relation between grade/condition/yr\_built and yr\_renovated?

Otherwise tree method

## NaNs filling in sqft\_basement

Perfect linear relation between  $(\text{sqft\_living} - \text{sqft\_above})$  and  $\text{sqft\_basement}$

But not so good linear relation between  $\text{sqft\_living}$  and  $\text{sqft\_basement}$

We then replaced missing values in  $\text{sqft\_basement}$  having the other two measures.

# Hypothesis

```
# H1: Properties located at specific geographic coordinates (latitude and longitude)
#     that are NOT waterfront are more likely to provide Smaller lot sizes.
# #### Turned out to be True!!!!
```

```
# H2: Properties that are characterized by either having at least
#     four bathrooms or having a larger living area than neighboring
#     properties have relatively higher grades.
# #### Turned out to be RELATIVELY TRUE!!!!
```

```
# H3: Properties that are historically older but in good condition are more likely to be
#     in original format and not been renovated, preserving their original architectural character.
# #### Turned out to be COMPLETELY TRUE!!!!
```







